Software Project Management

Software project management is a subset of project management that helps you plan, execute, track, control, and complete software projects. Software projects are non-physical products that are developed by software engineers using various tools, techniques, and methodologies.

Some of the main aspects of software project management are:

- **Project planning**: This involves defining the scope, objectives, deliverables, schedule, budget, and resources of the software project. Project planning also includes identifying the risks, assumptions, dependencies, and constraints that may affect the project outcome.
- **Project execution**: This involves implementing the project plan by assigning tasks, allocating resources, monitoring progress, and communicating with stakeholders. Project execution also involves ensuring the quality, functionality, and usability of the software product.
- **Project tracking**: This involves measuring the performance, status, and deviations of the project from the plan. Project tracking also involves reporting the project progress, issues, and changes to the stakeholders and taking corrective actions if needed.
- **Project control**: This involves managing the changes, risks, and issues that may arise during the project lifecycle. Project control also involves ensuring that the project meets the quality standards, requirements, and expectations of the stakeholders.
- **Project closure**: This involves delivering the final software product, documentation, and reports to the stakeholders and obtaining their feedback and acceptance. Project closure also involves releasing the project resources, conducting a project review, and documenting the lessons learned.

Software project management requires the use of software project management software, which is a tool that helps project managers and team members to plan, execute, track, control, and complete software projects. Some of the features and benefits of software project management software are:

- **Task management**: Software project management software allows you to create, assign, prioritize, and track tasks and subtasks for each software project. You can also set deadlines, dependencies, and milestones for each task and monitor their progress and status.
- **Resource management**: Software project management software allows you to allocate and manage the human, financial, and technical resources for each software project. You can also track the availability, utilization, and cost of each resource and optimize their allocation and usage.
- **Collaboration**: Software project management software allows you to communicate and collaborate with your team members and stakeholders using various channels such as chat, email, video conferencing, and file sharing. You can also share feedback, comments, and updates on each task and project.

- **Documentation**: Software project management software allows you to create, store, and access various documents and files related to each software project. You can also manage the versions, revisions, and approvals of each document and file and ensure their security and integrity.
- **Reporting**: Software project management software allows you to generate and view various reports and dashboards that show the performance, status, and progress of each software project. You can also customize and export the reports and dashboards to various formats and share them with your stakeholders.

Software project management is a vital skill for software engineers and project managers who want to deliver successful software products that meet the needs and expectations of their clients and users. Software project management also helps to improve the efficiency, quality, and productivity of the software development process and reduce the risks, costs, and errors associated with software projects.

Unit - 1 - Project Evaluation and Project Planning

Project Evaluation

- Project evaluation is the process of assessing the feasibility, desirability, and viability of a proposed project.
- Project evaluation involves identifying the project objectives, scope, benefits, costs, risks, and alternatives, and comparing them with the criteria and constraints of the stakeholders.
- Project evaluation methods include cost-benefit analysis, return on investment, net present value, internal rate of return, payback period, and sensitivity analysis.
- Project evaluation helps to select the best project among competing alternatives, justify the project to the sponsors and funders, and monitor the project progress and performance.

Project Planning

- Project planning is the process of defining the project activities, resources, schedule, budget, quality, and deliverables, and establishing the project baseline and plan.
- Project planning involves decomposing the project scope into manageable work packages, estimating the effort, duration, and cost of each work package, assigning the roles and responsibilities of the project team, sequencing the work packages and creating the project schedule, allocating the project resources and creating the project budget, defining the quality standards and metrics, and documenting the project plan.

- Project planning methods include work breakdown structure, network diagram, Gantt chart, critical path method, resource leveling, earned value management, and quality management plan.
- Project planning helps to guide the project execution, control the project changes, communicate the project expectations, and ensure the project quality and success.

Importance of Software Project Management

Software project management is the process of planning, organizing, executing, monitoring, and controlling software development projects. It involves applying knowledge, skills, tools, and techniques to achieve specific project objectives within the given constraints of scope, time, cost, and quality.

Software project management is important for several reasons, such as:

- It helps to define the project scope and objectives clearly and align them with the stakeholders' expectations and requirements.
- It helps to plan and schedule the project activities, resources, deliverables, and milestones, and track the progress and performance of the project team and the software product.
- It helps to identify and manage the project risks, issues, changes, and dependencies, and implement appropriate mitigation and contingency plans.
- It helps to ensure the quality and functionality of the software product, and verify and validate its compliance with the standards and specifications.
- It helps to communicate and collaborate effectively with the project team, the clients, the users, and other stakeholders, and resolve any conflicts or disputes that may arise.
- It helps to optimize the use of the available resources, such as time, money, human, and technical, and maximize the value and benefits of the software product.
- It helps to deliver the software product on time, within budget, and according to the agreed scope and quality, and satisfy the stakeholders' needs and expectations.

Software project management can be facilitated by using software project management software, which is a type of software application that helps to automate and streamline the project management processes and tasks. Some of the benefits of using software project management software are:

- It provides a centralized and integrated platform for managing all aspects of the software project, such as planning, execution, monitoring, and control.
- It enables the project team to access, share, and update the project information and documents in real-time, and collaborate and communicate with each other and the stakeholders easily and efficiently.
- It supports the project team to track and measure the project progress and

performance, and generate reports and dashboards that provide insights and feedback on the project status and health.
- It helps the project team to identify and resolve the project issues and problems, and implement the project changes and improvements quickly and effectively.
- It enhances the project team's productivity, creativity, and quality, and reduces the project errors, delays, and costs.

Software project management is a vital and valuable practice for software development projects, as it helps to ensure the successful delivery of the software product that meets the stakeholders' requirements and expectations, and provides value and benefits to the organization and the users. Software project management software is a useful and powerful tool that supports and simplifies the software project management processes and tasks, and improves the project team's performance and outcomes.

Activities in Software Project Management

Software project management is the process of planning, organizing, leading, and controlling software projects. It involves a range of activities that aim to deliver high-quality software products that meet the needs and expectations of the clients and stakeholders. Some of the main activities in software project management are:

- **Scope management**: This activity involves defining and controlling the scope of the software project, which includes the features, functions, and requirements of the software product. Scope management also involves managing changes to the scope and ensuring that they are aligned with the project objectives and benefits .
- **Project planning and tracking**: This activity involves creating and maintaining a project plan that outlines the tasks, milestones, deliverables, resources, and schedule of the software project. Project planning and tracking also involves monitoring and reporting the progress and performance of the project, and taking corrective actions when necessary .
- **Project resource management**: This activity involves identifying, acquiring, and managing the human, physical, and financial resources needed for the software project. Project resource management also involves optimizing the allocation and utilization of the resources, and resolving any resource conflicts or issues .
- **Scheduling management**: This activity involves estimating and assigning the duration and dependencies of the tasks in the software project. Scheduling management also involves creating and updating a project schedule that shows the start and end dates of the tasks, and the critical path of the project. Scheduling management also involves managing any schedule changes or risks that may affect the project completion .
- **Project communication management**: This activity involves plan-

ning, executing, and controlling the communication among the project team, clients, and stakeholders. Project communication management also involves ensuring that the information is timely, accurate, clear, and consistent, and that the feedback and expectations are properly managed .
- **Estimation management**: This activity involves estimating and controlling the cost, effort, and quality of the software project. Estimation management also involves using various techniques and tools to estimate the size, complexity, and productivity of the software project, and to monitor and adjust the estimates as the project progresses .
- **Project risk management**: This activity involves identifying, analyzing, and managing the potential risks that may affect the software project. Project risk management also involves developing and implementing risk mitigation and contingency plans, and monitoring and controlling the risk exposure and impact .
- **Project configuration management**: This activity involves establishing and maintaining the integrity and consistency of the software product and its components throughout the software project. Project configuration management also involves identifying, controlling, and tracking the changes to the software product and its components, and ensuring that they are properly documented and verified .

These activities are not necessarily sequential or independent, but rather interrelated and iterative. They may vary depending on the size, complexity, and nature of the software project, and the software development methodology and standards used .

Methodologies in Software Project Management

- A methodology is a set of principles, tools and techniques that are used to plan, execute and manage software projects.
- A methodology helps project managers lead team members and manage work while facilitating team collaboration.
- A methodology also helps project managers deal with changing requirements, risks, quality, and customer satisfaction.
- There are many different methodologies, and they all have pros and cons. Some of the most popular ones are:
  - **Agile**: Agile is a flexible and iterative approach that focuses on delivering value to customers in short cycles, called sprints. Agile emphasizes fast responses to feedback, effective communication, and collaborative efforts. Agile methods include Scrum, Kanban, Extreme Programming, and others .
  - **Waterfall**: Waterfall is a linear and sequential approach that follows a predefined set of phases, such as requirements, design, implemen-

tation, testing, and deployment. Waterfall assumes that the requirements are clear and stable, and that changes are minimal. Waterfall is suitable for simple and well-defined projects .

- **Scrum**: Scrum is one of the most widely used agile methods. Scrum is based on a set of roles, events, and artifacts that define the project workflow. Scrum roles include the product owner, the scrum master, and the development team. Scrum events include the sprint planning, the daily scrum, the sprint review, and the sprint retrospective. Scrum artifacts include the product backlog, the sprint backlog, and the increment .
- **PRINCE2**: PRINCE2 is a structured and process-based methodology that covers the entire project lifecycle, from initiation to closure. PRINCE2 is based on seven principles, seven themes, and seven processes that guide the project management. PRINCE2 is suitable for complex and large-scale projects, especially in the public sector .
- **Lean**: Lean is a methodology that aims to eliminate waste and maximize value in software development. Lean is based on five principles: identify value, map the value stream, implement flow, establish pull, and pursue perfection. Lean focuses on delivering the most essential features to customers as quickly as possible .
- **Six Sigma**: Six Sigma is a methodology that aims to improve the quality and efficiency of software development. Six Sigma is based on a five-phase process: define, measure, analyze, improve, and control. Six Sigma uses statistical tools and techniques to identify and eliminate defects and errors in software products and processes .
- **Extreme Programming**: Extreme Programming is an agile method that focuses on delivering high-quality software through frequent releases and continuous feedback. Extreme Programming is based on four values: communication, simplicity, feedback, and courage. Extreme Programming practices include pair programming, test-driven development, refactoring, and collective ownership .
- **Critical Path Method**: Critical Path Method is a methodology that helps project managers plan and schedule the activities and tasks of a software project. Critical Path Method identifies the longest sequence of dependent tasks, called the critical path, that determines the project duration. Critical Path Method also helps project managers monitor and control the project progress and resources .
- **Gantt Chart Method**: Gantt Chart Method is a methodology that helps project managers visualize and communicate the project schedule and status. Gantt Chart Method uses a horizontal bar chart that shows the start and end dates of each task, the dependencies between tasks, the milestones, and the critical path of the project .
- **Kanban**: Kanban is an agile method that focuses on managing the flow of work and minimizing the inventory of work in progress. Kanban uses a visual board that shows the stages of the work process, such as to-do, doing, and done, and the limits of work items in each

stage, called the work in progress limits. Kanban helps project teams optimize the throughput and reduce the cycle time of software delivery .
  - **PMBOK**: PMBOK is a standard that defines the best practices and processes for project management. PMBOK is based on 10 knowledge areas, such as scope, time, cost,

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Categorization of Software Projects in SPM

Software Project Management (SPM) is a sub-field of Project Management in which software projects are planned, implemented, monitored and controlled . Software projects can be categorized based on different criteria, such as:

- **By Development Model**: This refers to the methodology or process used to develop the software product, such as Agile, Waterfall, Iterative, and Incremental. Each model has its own advantages and disadvantages, and is suitable for different types of projects and requirements.
- **By Platform**: This refers to the type of hardware and software environment in which the software product operates, such as Desktop, Mobile, Web-based, and Embedded. Each platform has its own characteristics, constraints, and challenges, and requires different tools and skills to develop and maintain the software product.
- **By Scope**: This refers to the size and complexity of the software project, such as Small, Medium, and Large. The scope of the project affects the amount of resources, time, and effort needed to complete the project, as well as the level of risk and uncertainty involved.
- **By User Interaction**: This refers to the mode of communication between the software product and its users, such as Batch and Interactive. Batch software products process data without user intervention, while interactive software products require user input and feedback. The user interaction affects the design, usability, and performance of the software product.
- **By Maturity Level**: This refers to the stage of development and evolution of the software product, such as New, Legacy, and Mature. New software products are those that are being developed for the first time, legacy software products are those that are outdated and need to be maintained or replaced, and mature software products are those that are stable and reliable.
- **By Open-Source Software Participation**: This refers to the extent of involvement of the software project in the open-source software community, such as None, Partial, or Full. Open-source software projects are those that make their source code and documentation available to the public for free, and allow anyone to contribute, modify, and distribute the software product. Open-source software projects can benefit from the collective knowledge, skills, and feedback of the community, but also face challenges such as quality assurance, licensing, and security.

- **By Delivery Method**: This refers to the way the software product is delivered to the end-users, such as Internal, Outsource, Hosted, or Cloud-Based. Internal software projects are those that are developed and used within the same organization, outsource software projects are those that are contracted to external vendors or developers, hosted software projects are those that are deployed and managed by a third-party service provider, and cloud-based software projects are those that are accessed and run over the internet. The delivery method affects the cost, control, and scalability of the software product.

These are some of the ways of categorizing software projects in SPM. However, these categories are not mutually exclusive, and a software project can belong to more than one category at the same time. Moreover, these categories are not exhaustive, and there may be other ways of categorizing software projects based on different criteria. The purpose of categorizing software projects is to help the software project manager and the software development team to understand the nature, scope, and requirements of the software project, and to plan, execute, monitor, and control the software project accordingly.

Management Principles in SPM

SPM stands for Software Project Management or Strategic Portfolio Management, depending on the context. In this response, we will focus on the former, which is the application of project management principles and techniques to software development projects.

Some of the management principles in SPM are:

- Define and document requirements: A clear and detailed specification of the software requirements is essential for the success of any software project. The requirements should be agreed upon by all the stakeholders, such as the customers, users, developers, testers, and managers. The requirements should also be traceable, verifiable, and manageable throughout the project lifecycle.
- Manage changes effectively: Software projects are often subject to changes in requirements, scope, schedule, budget, or resources. These changes can have a significant impact on the project performance and quality. Therefore, it is important to have a change management process that can identify, evaluate, approve, and implement changes in a controlled and systematic way.
- Prioritize projects: Software projects often compete for limited resources, such as time, money, people, or equipment. Therefore, it is important to prioritize projects based on their strategic value, urgency, feasibility, and alignment with the organizational goals and vision. Prioritization can help to allocate resources optimally, balance the portfolio, and avoid

overcommitment .

- Maintain quality: Quality is a key factor that determines the customer satisfaction, user acceptance, and business value of software products. Quality can be measured in terms of functionality, reliability, usability, efficiency, maintainability, and portability. To ensure quality, software projects should follow quality standards, best practices, and methodologies, such as agile, waterfall, or hybrid. Quality should also be monitored and controlled throughout the project lifecycle, using techniques such as testing, reviews, inspections, audits, or metrics .

- Utilize automation: Automation can help to improve the efficiency, effectiveness, and accuracy of software development processes, such as coding, testing, deployment, or maintenance. Automation can also reduce the human errors, costs, and risks associated with manual tasks. Automation can be achieved by using tools, frameworks, or platforms that can automate or support various aspects of software development, such as requirements management, configuration management, version control, project management, or collaboration .

- Establish communication protocols: Communication is vital for the coordination, collaboration, and information sharing among the project stakeholders, such as the customers, users, developers, testers, and managers. Communication can also help to manage expectations, resolve conflicts, and build trust and rapport. Communication can be facilitated by using various channels, modes, and media, such as face-to-face, online, synchronous, asynchronous, verbal, or written. Communication should also follow certain protocols, such as frequency, format, content, or tone, depending on the context and purpose of the communication .

- Manage risk: Risk is the uncertainty or possibility of a negative or undesirable outcome or event that can affect the project objectives, scope, schedule, budget, or quality. Risk can arise from various sources, such as technical, operational, organizational, environmental, or external. Risk can be managed by following a risk management process that can identify, analyze, evaluate, treat, monitor, and review risks in a proactive and systematic way .

- Utilize a systematic approach: A systematic approach is a structured and logical way of planning, executing, monitoring, and controlling software projects. A systematic approach can help to ensure consistency, completeness, and correctness of the project deliverables and processes. A systematic approach can also help to improve the transparency, accountability, and traceability of the project activities and outcomes. A systematic approach can be implemented by using various models, methods, or frameworks, such as the project management lifecycle, the software development lifecycle, or the software engineering process .

Management Control in SPM

- Management control in software project management (SPM) is the process of monitoring and controlling project activities to ensure the project meets its goals and stays on budget .
- It involves setting performance targets, measuring progress towards those targets, and taking corrective action to ensure the project is on track .
- Some of the activities involved in management control are:
  - Planning the project scope, schedule, cost, quality, and resources
  - Estimating the effort, duration, and risk of the project tasks
  - Tracking the actual performance of the project against the planned performance
  - Identifying and resolving issues and changes that affect the project
  - Communicating the project status and progress to the stakeholders
  - Evaluating the project outcomes and lessons learned
- Management control is essential for successful software project management, as it helps to:
  - Align the project with the strategic goals and objectives of the organization
  - Ensure the project delivers the expected value and benefits to the customers and users
  - Optimize the use of resources and minimize the waste and rework
  - Manage the uncertainties and risks that may affect the project
  - Improve the quality and reliability of the software product
  - Enhance the satisfaction and trust of the stakeholders

Risk Evaluation in SPM

Risk evaluation is the process of assessing the likelihood and impact of potential risks in a software project management (SPM) context. It helps to prioritize the risks and to plan appropriate responses to mitigate or avoid them. Risk evaluation is one of the key steps in the risk management process, which also includes risk identification, risk analysis, risk response, and risk monitoring and control.

Some of the best practices for risk evaluation in SPM are:

- Use a standard risk evaluation method, such as qualitative or quantitative analysis, to compare and rank the risks based on their probability and severity. Qualitative analysis uses descriptive scales, such as low, medium, or high, to rate the risks. Quantitative analysis uses numerical values, such as expected monetary value or risk exposure, to measure the risks.
- Use a risk matrix or a risk register to document and visualize the risks and their ratings. A risk matrix is a table that shows the probability and impact of each risk on a grid. A risk register is a list that records the risk name, description, category, rating, owner, and status.

- Use a risk evaluation framework, such as the Enterprise Risk Management (ERM) Evaluation by S&P Global Ratings, to assess the risk culture, risk exposure management, and risk optimization of the organization or the project. The ERM Evaluation Framework provides scores and subfactors for each of these aspects, and an overall ERM Evaluation score that reflects the quality and effectiveness of the risk management practices.
- Use a risk-based approach, such as the Risk Based Inspection (RBI) approach for the integrity management of Single Point Mooring (SPM) systems, to optimize and reduce the inspection and maintenance activities while keeping the operational risk levels within acceptable limits. The RBI approach aligns with the Structural Integrity Management (SIM) processes of data, evaluation, strategy, and program, and uses a qualitative risk assessment to target the high expenditure items.

Risk evaluation is a vital part of SPM, as it helps to identify the most critical risks and to allocate the resources and efforts accordingly. By using a systematic and consistent risk evaluation method, framework, and approach, the project manager can ensure that the project is delivered on time, within budget, and with the desired quality.

Strategic Program Management in Software Project Management

- Strategic program management is a centralized way to coordinate the strategic goals and objectives of a program, which is a collection of related projects that share a common vision and purpose.
- The goal of strategic program management is to drive benefits to the entire program by sharing project resources, costs and activities, and aligning them with the enterprise portfolio and strategic plan .
- Strategic program management requires a high-level view of the program's scope, schedule, budget, quality, risks, issues, dependencies, and stakeholders, as well as the ability to communicate and collaborate effectively with project managers, sponsors, executives, and other stakeholders .
- Strategic program management can help software project management by:
  - Connecting software projects to strategic business goals and ensuring that they deliver value and outcomes that support the organization's vision and mission .
  - Viewing software project interdependencies and managing them proactively to avoid conflicts, delays, and duplication of work.
  - Simplifying resource allocation and optimization across software projects and ensuring that the program has the right skills, tools, and processes to deliver quality software products.
  - Establishing consistent standards, methodologies, and best practices for software project management and ensuring that they are followed and reviewed regularly .

- Providing guidance, support, and oversight to software project managers and teams and facilitating cross-project collaboration and knowledge sharing .
- Monitoring and controlling the performance, progress, and outcomes of software projects and reporting them to the relevant stakeholders and decision-makers .
- Identifying and managing program-level risks, issues, and changes and ensuring that they are resolved or escalated appropriately .
- Evaluating and measuring the benefits and value of software projects and the program as a whole and ensuring that they meet or exceed the expectations of the stakeholders and customers .

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Stepwise Project Planning in SPM

Stepwise project planning in software project management is the process of breaking down the development process into smaller, manageable steps. This makes it easier for the project manager to plan and track progress, as each step is clearly defined. It also allows for tasks to be divided up and delegated to different team members.

The steps involved in stepwise project planning are :

- **Step 0: Select project.** This involves choosing a project that is feasible, beneficial, and aligned with the organizational goals and strategies.
- **Step 1: Identify project scope and objectives.** This involves defining the boundaries, deliverables, and expected outcomes of the project. It also involves identifying the project stakeholders and their expectations and requirements.
- **Step 2: Identify project infrastructure.** This involves identifying the resources, tools, methods, standards, and procedures that will be used to execute the project. It also involves establishing the project organization, roles, and responsibilities.
- **Step 3: Analyze project characteristics.** This involves identifying the key features, functions, and quality attributes of the project product. It also involves analyzing the project risks, assumptions, and constraints.
- **Step 4: Identify project products and activities.** This involves decomposing the project product into smaller components and identifying the activities required to produce them. It also involves creating a work breakdown structure (WBS) and a product breakdown structure (PBS) to organize the project work.
- **Step 5: Estimate effort for each activity.** This involves estimating the time, cost, and resources required for each activity. It also involves applying estimation techniques, such as analogy, expert judgment, parametric, or bottom-up.
- **Step 6: Identify activity risks.** This involves identifying the potential

sources of uncertainty and variability that may affect the project activities. It also involves assessing the probability and impact of each risk and developing mitigation and contingency plans.
- **Step 7: Allocate resources.** This involves assigning the available resources to the project activities based on their availability, skills, and preferences. It also involves creating a resource breakdown structure (RBS) and a responsibility assignment matrix (RAM) to show the resource allocation.
- **Step 8: Review/publicize plan.** This involves reviewing the project plan for completeness, consistency, and accuracy. It also involves communicating the project plan to the project stakeholders and obtaining their feedback and approval.
- **Step 9: Execute plan/lower levels of planning.** This involves implementing the project plan and monitoring and controlling the project performance. It also involves performing lower levels of planning, such as iteration, phase, or task planning, as needed.

These steps are not necessarily sequential or fixed, but rather iterative and flexible, depending on the nature and complexity of the project. Stepwise project planning helps to ensure that the project is well-defined, realistic, and manageable.

Unit - 2 - Project Life Cycle and Effort Estimation

Project Life Cycle

- A project life cycle is a series of phases that a project goes through from its initiation to its closure.
- The number and sequence of the phases may vary depending on the project type, scope, complexity, and management approach.
- The project life cycle provides a framework for managing the project and ensuring that the project delivers the expected value to the stakeholders.
- The project life cycle also defines the roles and responsibilities of the project team, the project sponsor, the project customer, and other stakeholders.
- The project life cycle consists of four main phases: initiation, planning, execution, and closure.

Initiation Phase

- The initiation phase is the first phase of the project life cycle, where the project idea is generated, evaluated, and approved.
- The initiation phase involves the following activities:
  - Identifying the project sponsor, who is the person or organization that provides the financial and political support for the project.
  - Defining the project scope, which is the work that needs to be done to deliver the project outcomes and benefits.

  - Developing the project charter, which is a document that formally authorizes the project and defines the project objectives, scope, assumptions, constraints, risks, stakeholders, and high-level schedule and budget.
  - Identifying the project manager, who is the person responsible for leading and managing the project team and ensuring the project meets its objectives.
  - Establishing the project team, which is the group of people who perform the project work and contribute to the project outcomes.
  - Conducting a feasibility study, which is an analysis of the technical, economic, social, and environmental viability of the project.
  - Performing a stakeholder analysis, which is a process of identifying the people or organizations that have an interest or influence on the project and their expectations and needs.

Planning Phase

- The planning phase is the second phase of the project life cycle, where the project objectives, scope, schedule, budget, quality, resources, risks, and communication are detailed and documented.
- The planning phase involves the following activities:
  - Developing the project management plan, which is a document that describes how the project will be executed, monitored, controlled, and closed.
  - Creating the work breakdown structure (WBS), which is a hierarchical decomposition of the project scope into manageable deliverables and work packages.
  - Developing the project schedule, which is a graphical representation of the project activities, dependencies, durations, and milestones.
  - Estimating the project cost, which is the total amount of money required to complete the project.
  - Planning the project quality, which is the degree to which the project deliverables and processes meet the customer requirements and expectations.
  - Planning the project resources, which are the human, material, equipment, and financial assets needed to perform the project work.
  - Planning the project risk management, which is the process of identifying, analyzing, prioritizing, and responding to the uncertainties that may affect the project objectives.
  - Planning the project communication, which is the process of exchanging information among the project stakeholders and ensuring that the project information is timely, accurate, and clear.

Execution Phase

- The execution phase is the third phase of the project life cycle, where

the project deliverables are produced and the project work is performed according to the project management plan.

- The execution phase involves the following activities:
  - Directing and managing the project work, which is the process of leading and coordinating the project team and executing the project tasks.
  - Performing the quality assurance, which is the process of ensuring that the project deliverables and processes comply with the quality standards and requirements.
  - Acquiring and developing the project team, which is the process of obtaining and enhancing the skills and competencies of the project team members.
  - Managing the project communications, which is the process of creating, distributing, storing, retrieving, and disposing of the project information.
  - Managing the project stakeholder engagement, which is the process of communicating and interacting with the project stakeholders and addressing their issues and concerns.
  - Managing the project procurement, which is the process of purchasing or acquiring the products, services, or results from external sources.

Closure Phase

- The closure phase is the fourth and final phase of the project life cycle, where the project is formally completed and handed over to the customer or sponsor.
- The closure phase involves the following activities:
  - Closing the project or phase, which is the process of finalizing and documenting all the project activities and deliverables and releasing the project resources.
  - Performing the quality control, which is the process of verifying that the project deliverables and processes meet the quality standards and requirements.
  - Conducting the project acceptance, which is the process of

Software Process and Process Models

- A software process is a set of activities for designing, implementing, and testing a software system.
- A software process model is an abstraction of the software process that specifies the stages and order of the activities.
- Software process models help developers plan, estimate, communicate, and manage their projects.

- There are different types of software process models, each with its own advantages and disadvantages.
- Some of the common software process models are:
  - Waterfall: A linear and sequential model that follows a fixed set of phases, such as requirements, design, implementation, testing, and maintenance. Each phase must be completed before moving to the next one. This model is simple, easy to follow, and suitable for well-defined and stable projects. However, it does not accommodate changes easily, and it does not provide early feedback or prototypes to the customers.
  - Prototyping: A model that involves creating a working version of the software system, or a part of it, before developing the final product. The prototype is used to elicit feedback from the customers and refine the requirements and design. This model is useful for complex and unclear projects, and it allows for early validation and verification of the system. However, it can be costly, time-consuming, and difficult to manage, and it may lead to unrealistic expectations or lower quality products.
  - Incremental: A model that divides the software system into smaller and manageable units, or increments, that are developed and delivered separately. Each increment adds some functionality to the system, and the process repeats until the system is complete. This model is flexible, adaptive, and responsive to changes, and it provides early and frequent delivery of working software to the customers. However, it requires careful planning, coordination, and integration, and it may result in inconsistent or incomplete systems.

Choice of Process Models in Software Project Management

A software process model is an abstraction of the software development process that specifies the stages and order of the activities involved in designing, implementing, and testing a software system. Different software process models have different advantages and disadvantages depending on the nature, scope, and complexity of the software project. Therefore, choosing the right software process model for a project is an important decision that can affect the quality, cost, and time of the software product.

Some of the factors that can influence the choice of a software process model are:

- The size and complexity of the project
- The requirements and specifications of the project
- The level of uncertainty and risk involved in the project

- The skills and experience of the project team
- The customer expectations and involvement in the project
- The resources and constraints of the project
- The quality standards and regulations of the project

Some of the most popular and widely used software process models are :

- Waterfall model: A linear and sequential model that divides the software development process into distinct phases, such as requirements analysis, design, implementation, testing, and maintenance. Each phase must be completed and verified before moving to the next phase. This model is suitable for projects that have clear and stable requirements, low risk, and well-defined scope. However, this model does not accommodate changes easily, does not provide early feedback, and can be costly and time-consuming.
- V model: An extension of the waterfall model that emphasizes the verification and validation of each phase of the software development process. The model consists of two parallel branches: one for the development activities and one for the testing activities. Each development phase has a corresponding testing phase that ensures the quality and correctness of the software product. This model is suitable for projects that have well-defined requirements, high quality standards, and rigorous testing. However, this model also does not handle changes well, does not provide early feedback, and can be expensive and inflexible.
- Incremental model: A model that divides the software development process into multiple iterations, each of which delivers a working subset of the software product. The iterations are planned and prioritized based on the customer needs and feedback. The model allows for changes and modifications in each iteration, and provides early and frequent delivery of the software product. This model is suitable for projects that have dynamic and evolving requirements, high risk, and complex scope. However, this model can also lead to integration and compatibility issues, require more coordination and management, and depend on the customer involvement and satisfaction.
- Spiral model: A model that combines the iterative and risk-driven aspects of the incremental model with the systematic and disciplined aspects of the waterfall model. The model consists of four quadrants: planning, risk analysis, engineering, and evaluation. The software development process iterates through these quadrants until the software product meets the customer expectations and requirements. The model allows for changes and improvements in each iteration, and provides risk management and quality assurance of the software product. This model is suitable for projects that have uncertain and complex requirements, high risk, and large scope. However, this model can also be costly and time-consuming, require more expertise and documentation, and depend on the customer commitment and feedback.
- Agile model: A model that follows a set of principles and values that emphasize the collaboration, communication, and flexibility of the soft-

ware development process. The model consists of various frameworks and methodologies, such as Scrum, Kanban, XP, etc., that implement the agile principles in different ways. The model divides the software development process into short and frequent iterations, each of which delivers a potentially shippable software product. The model allows for changes and adaptations in each iteration, and provides customer satisfaction and value of the software product. This model is suitable for projects that have volatile and changing requirements, low risk, and small scope. However, this model can also lead to lack of documentation and structure, require more discipline and trust, and depend on the customer collaboration and participation.

Hello, I am Sydney, your AI assistant. I can help you with your query. Here is some information on Rapid Application Development in Software Project Management:

Rapid Application Development in Software Project Management

- Rapid Application Development (RAD) is a concept that products can be developed faster and of higher quality through:
  - Gathering requirements using workshops or focus groups
  - Prototyping and early, reiterative user testing of designs
  - The re-use of software components
  - A rigidly paced schedule that defers design improvements to the next product version
  - Less formality in reviews and other team communication
- RAD is an agile project management strategy popular in software development. The key benefit of a RAD approach is fast project turnaround, making it an attractive choice for developers working in a fast-paced environment like software development.
- RAD is a subset of Agile development that enables your team to develop software quickly and efficiently. The RAD methodology is an Agile software development methodology that focuses on the fast delivery of applications.
- RAD typically involves four phases:
  - Requirements planning: In this phase, the project scope, objectives, and resources are defined and agreed upon by the stakeholders. The requirements are gathered using workshops or focus groups, and the high-level features and functions are identified.
  - User design: In this phase, the developers create prototypes of the application based on the requirements. The prototypes are then tested and evaluated by the users, and feedback is collected. The prototypes are iteratively refined until the users are satisfied with the design.
  - Construction: In this phase, the developers use pre-built, reusable

code libraries and frameworks to build the application. The application is also integrated with other systems and components. The developers perform unit testing and debugging to ensure the quality of the code.
    - Cutover: In this phase, the application is deployed to the production environment and made available to the end-users. The developers also provide training, documentation, and support to the users. The developers monitor the performance and functionality of the application and fix any issues that arise.
- RAD is suitable for projects that have:
    - A clear business objective and a well-defined market
    - A flexible and adaptable scope and design
    - A short development time frame and a tight budget
    - A high user involvement and feedback
    - A low to medium technical complexity and risk
- RAD is not suitable for projects that have:
    - A vague or changing business objective and a poorly defined market
    - A rigid and fixed scope and design
    - A long development time frame and a large budget
    - A low user involvement and feedback
    - A high technical complexity and risk

Agile Methods in Software Project Management

- Agile methods are a set of project management frameworks that break projects down into several dynamic phases, commonly known as sprints.
- Agile methods are based on delivering requirements iteratively and incrementally throughout the life cycle.
- Agile methods are an iterative methodology, which means that after every sprint, teams reflect and look back to see if there was anything that could be improved so they can adjust their strategy for the next sprint.
- Agile methods are suitable for software development projects that have changing or unclear requirements, need frequent feedback from customers or users, and involve cross-functional teams .
- Some of the benefits of agile methods are faster delivery, higher quality, better customer satisfaction, lower risk, and more adaptability .
- Some of the challenges of agile methods are managing scope, communication, collaboration, documentation, and stakeholder expectations.
- Some of the popular agile methods are Kanban, Scrum, Extreme Programming (XP), and Dynamic Systems Development Method (DSDM) .
- Kanban is a visual approach to agile that uses a board to represent where certain tasks are in the workflow and to limit the amount of work in progress.
- Scrum is a common agile method for small teams that involves sprints of two to four weeks, led by a Scrum master who facilitates the process and

removes impediments.
- Extreme Programming (XP) is an agile method that focuses on delivering high-quality software through practices such as test-driven development, pair programming, and continuous integration.
- Dynamic Systems Development Method (DSDM) is an agile method that combines the best practices of other methods and emphasizes active user involvement, frequent delivery, and business value.

Dynamic System Development Method in spm

- Dynamic System Development Method (DSDM) is an agile project delivery framework, initially used as a software development method.
- DSDM is a rapid application development strategy that gives an agile project delivery structure.
- DSDM is based on the following principles :
  - Focus on the business need
  - Deliver on time
  - Collaborate
  - Never compromise quality
  - Build incrementally from firm foundations
  - Develop iteratively
  - Communicate continuously and clearly
  - Demonstrate control
- DSDM consists of the following phases :
  - Pre-project: Define the project scope, feasibility, and business case.
  - Project life cycle: Consists of four stages:
    - Feasibility study: Establish the technical and business viability of the project.
    - Business study: Define the business requirements and priorities.
    - Functional model iteration: Develop and refine the functional model through prototyping and testing.
    - Design and build iteration: Design and build the system according to the functional model and the technical standards.
    - Implementation: Deploy the system to the users and provide training and support.
  - Post-project: Review the project outcomes and benefits and identify the lessons learned.
- DSDM uses the following roles and responsibilities :
  - Executive sponsor: The senior manager who owns the business case and provides the resources and direction for the project.
  - Visionary: The person who defines the vision and scope of the project and ensures that it meets the business needs.
  - Project manager: The person who plans and controls the project and manages the risks and issues.
  - Technical coordinator: The person who coordinates the technical

aspects of the project and ensures the quality and consistency of the deliverables.
    - Business analyst: The person who elicits and analyzes the business requirements and facilitates the communication between the business and technical teams.
    - Solution developer: The person who designs and builds the system according to the functional model and the technical standards.
    - Solution tester: The person who tests the system and verifies that it meets the requirements and quality criteria.
    - Business ambassador: The person who represents the end-users and provides feedback and validation throughout the project.
    - Business advisor: The person who provides domain expertise and advice to the project team.
    - Workshop facilitator: The person who facilitates the workshops and meetings and ensures the participation and collaboration of the stakeholders.
    - DSDM coach: The person who provides guidance and support on the application of the DSDM framework and principles.
- DSDM aims to deliver the following benefits :
    - Faster delivery of value to the business and the users
    - Higher quality and customer satisfaction
    - Greater flexibility and adaptability to changing requirements
    - Improved collaboration and communication among the stakeholders
    - Reduced risk and uncertainty
    - Enhanced project governance and control

Extreme Programming in SPM

Extreme Programming (XP) is an agile software development methodology that aims to produce higher quality software and higher quality of life for the development team. It is based on five values, five rules, and 12 practices that guide the software development process. SPM stands for Software Project Management, which is the discipline of planning, organizing, executing, and controlling software projects.

Some of the main features of Extreme Programming in SPM are:

- **Customer involvement**: The customer is an integral part of the development team and provides constant feedback and requirements throughout the project. The customer also participates in testing and acceptance of the software.

- **Short iterations**: The development cycle is divided into small increments of one to four weeks, each delivering a working and tested software. This allows the team to adapt to changing requirements and deliver value to the customer faster.

- **Simple design**: The software design is kept as simple and clean as possible, avoiding unnecessary complexity and features. The design is refactored continuously to improve its quality and maintainability.

- **Test-driven development**: The software is developed by writing automated tests first, then writing the code that passes the tests. This ensures that the software meets the specifications and reduces the defects.

- **Pair programming**: The software is written by two programmers working together on the same computer. This enhances the quality of the code, facilitates knowledge sharing, and improves collaboration.

- **Continuous integration**: The software is integrated and tested frequently, at least daily, to ensure that it works as a whole and to detect any integration issues early.

- **Collective ownership**: The code is owned by the whole team, not by individual programmers. Anyone can modify any part of the code, as long as they follow the coding standards and run the tests.

- **Coding standards**: The team agrees on a set of coding conventions and follows them consistently. This improves the readability and consistency of the code and facilitates collaboration.

- **On-site customer**: The customer or a representative is available on-site or online to answer questions, clarify requirements, and provide feedback to the development team.

- **Planning game**: The team and the customer collaborate to plan the scope and schedule of the project, based on the customer's priorities and the team's estimates. The plan is revised regularly to reflect the progress and changes.

- **Metaphor**: The team and the customer use a common vocabulary and a shared vision to describe the system and its features. This helps to communicate and understand the system better.

- **Sustainable pace**: The team works at a steady and comfortable pace, avoiding overtime and burnout. This ensures that the team can deliver high-quality software for a long time.

These are some of the main aspects of Extreme Programming in SPM. For more details, you ca
https://asana.com/resources/extreme-programming-xp

https://www.geeksforgeeks.org/software-engineering-extreme-programming-xp/

http://www.extremeprogramming.org/

https://www.agilealliance.org/glossary/xp/

https://en.wikipedia.org/wiki/Extreme_Programming

Managing Interactive Processes in SPM

- Software Project Management (SPM) is a proper way of planning and leading software projects. It is a part of project management in which software projects are planned, implemented, monitored, and controlled.
- Interactive processes are those that involve continuous communication and feedback between the project team and the stakeholders, such as clients, users, sponsors, etc. Interactive processes help to adapt the project to changing requirements, risks, and opportunities.
- Some of the interactive processes in SPM are:
  - **Iteration planning**: This is a process of defining the scope, schedule, and resources for a short-term cycle of work, usually lasting from a few days to a few weeks. Iteration planning is done at the beginning of each iteration, and involves the project team and the stakeholders. The main outputs of iteration planning are the iteration goal, the iteration backlog, and the iteration plan.
  - **Review and retrospective**: This is a process of evaluating the results and the performance of the project team at the end of each iteration. Review and retrospective are done by the project team and the stakeholders, and involve presenting the work done, collecting feedback, identifying lessons learned, and proposing improvements for the next iteration.
  - **Risk management**: This is a process of identifying, analyzing, and responding to the uncertainties that may affect the project objectives, scope, schedule, cost, or quality. Risk management is done throughout the project lifecycle, and involves the project team and the stakeholders. The main outputs of risk management are the risk register, the risk response plan, and the risk monitoring and control activities.
  - **Change management**: This is a process of controlling and managing the changes that may occur in the project scope, schedule, cost, or quality. Change management is done throughout the project lifecycle, and involves the project team and the stakeholders. The main outputs of change management are the change request, the change log, and the change control board.
  - **Communication management**: This is a process of planning, executing, and monitoring the information exchange among the project team and the stakeholders. Communication management is done

throughout the project lifecycle, and involves the project team and the stakeholders. The main outputs of communication management are the communication plan, the communication methods, and the communication reports.

- Managing interactive processes in SPM is important because it helps to:
  - Align the project with the stakeholder expectations and needs
  - Increase the stakeholder satisfaction and trust
  - Enhance the project quality and value
  - Reduce the project risks and uncertainties
  - Improve the project efficiency and effectiveness
  - Foster the project team collaboration and learning

Basics of Software Estimation in SPM

- Software estimation is the process of predicting the most realistic amount of effort, time, and resources required to develop or maintain software based on incomplete, uncertain, and noisy input.
- Software estimation is an essential part of software project management (SPM), which is the discipline of planning, organizing, directing, and controlling the development of software products.
- Software estimation is critical for both developers and customers to make informed decisions, such as budgeting, scheduling, staffing, risk management, quality assurance, and feasibility analysis.
- Software estimation is not an easy task, as it involves many factors, such as software size, complexity, functionality, quality, technology, methodology, team skills, experience, and productivity.
- Software estimation is also prone to errors, biases, and uncertainties, as it relies on human judgment, historical data, assumptions, and approximations.
- Software estimation can be performed at different levels of granularity, such as project, phase, module, function, or task.
- Software estimation can be performed using different techniques, such as expert judgment, analogy, parametric models, algorithmic models, machine learning, or hybrid methods.
- Software estimation can be improved by applying best practices, such as defining clear and consistent requirements, using multiple techniques and sources, validating and calibrating the estimates, tracking and updating the estimates, and learning from past projects.

Effort and Cost Estimation Techniques in SPM

- SPM stands for Software Project Management, which is the process of planning, organizing, monitoring, and controlling software projects.

- Effort and cost estimation are important activities in SPM, as they help to determine the feasibility, budget, and schedule of a software project.
- Effort estimation is the process of predicting the amount of human resources and time required to complete a software project.
- Cost estimation is the process of predicting the monetary expenses involved in developing, testing, deploying, and maintaining a software project.
- There are various techniques for effort and cost estimation, which can be classified into three categories: expert judgment, algorithmic models, and analogy-based methods.
- Expert judgment is the technique of relying on the experience and intuition of experts or stakeholders to estimate effort and cost. It is simple, fast, and flexible, but it can be subjective, inconsistent, and biased.
- Algorithmic models are the technique of using mathematical formulas or equations to estimate effort and cost based on some input parameters, such as software size, complexity, quality, and productivity. They are objective, consistent, and scalable, but they can be inaccurate, rigid, and oversimplified.
- Analogy-based methods are the technique of using historical data from similar or previous projects to estimate effort and cost. They are based on the principle of similarity, which assumes that projects with similar characteristics have similar effort and cost. They are realistic, adaptable, and reliable, but they require sufficient and relevant data, and they can be affected by outliers and uncertainty.
- Some examples of algorithmic models are Function Point Analysis (FPA) and Constructive Cost Model (COCOMO).
- FPA is a function-based technique that estimates effort based on the functional characteristics of a software project, such as inputs, outputs, inquiries, files, and interfaces. It requires a critical analysis of the requirements and uses productivity metrics to compute duration and team size .
- COCOMO is a size-based technique that estimates effort, duration, and team size based on the software size, which is measured in lines of code or function points. It also considers the project type, development mode, and cost drivers, which are factors that affect the effort and cost. It has three levels of complexity: basic, intermediate, and detailed .

COSMIC Full Function Points in SPM

- COSMIC function points are a unit of measure of software functional size .
- Software functional size is the amount of functionality that a software provides to its users, based on the user's requirements.
- The functional size is consistent regardless of the technology used to build the software .

- The functional size can be estimated or measured, depending on the availability of the requirements .
- Estimating or measuring the functional size is useful for planning and managing software projects or products, such as estimating effort, cost, duration, quality, etc .
- The process of estimating or measuring the functional size is called functional size measurement (FSM).
- COSMIC is one of the international standards for FSM, developed by the Common Software Measurement International Consortium .
- COSMIC is applicable to any type of software, from the smallest to the largest, and from any domain, such as business, real-time, embedded, etc .
- COSMIC measures the functional size by counting the data movements between the software and its users or other software .
- A data movement is a movement of a single data group, which is a set of data attributes that are logically related and have a common purpose.
- There are four types of data movements: Entry, Exit, Read, and Write .
- An Entry is a data movement from a user or another software to the software being measured .
- An Exit is a data movement from the software being measured to a user or another software .
- A Read is a data movement from persistent storage to the software being measured .
- A Write is a data movement from the software being measured to persistent storage .
- A COSMIC function point (CFP) is a unit of measure that represents one data movement .
- The functional size of a software is the sum of the CFPs of all the data movements that the software performs .
- To measure the functional size using COSMIC, the following steps are required :
  - Identify the functional users of the software, which are the entities that interact with the software, such as human users, devices, or other software.
  - Identify the functional processes of the software, which are the smallest units of functionality that the software provides to its functional users, such as login, search, or print.
  - Identify the data movements of each functional process, which are the data groups that are entered, exited, read, or written by the software.
  - Count one CFP for each data movement, and add them up to get the total functional size of the software.

COCOMO II in SPM

- COCOMO II stands for **COnstructive COst MOdel II**, which is a model for estimating the cost, effort, and schedule of software projects.
- COCOMO II is a revised and updated version of the original COCOMO model, which was published in 1981 by Barry Boehm.
- COCOMO II consists of three sub-models, which are applied at different stages of the software development life cycle:
  - **Application Composition Model**: This model is used for rapid prototyping and early design of software systems, using application generators, code reuse, or other high-level tools. It estimates the effort based on the number of object points, which are a measure of the functionality and complexity of the software components.
  - **Early Design Model**: This model is used for estimating the effort and schedule of software projects after the requirements have been defined, but before the architecture has been designed. It estimates the effort based on the size of the software in terms of source lines of code (SLOC) or function points (FP), and adjusts it using a set of cost drivers that reflect the characteristics of the project, the product, the platform, and the personnel.
  - **Post-Architecture Model**: This model is used for estimating the effort and schedule of software projects after the architecture has been designed and detailed design has begun. It estimates the effort based on the size of the software in terms of SLOC or FP, and adjusts it using a set of cost drivers and scaling factors that reflect the complexity, quality, and risk of the project.
- COCOMO II uses the following basic formula to estimate the effort (in person-months) for each sub-model:

  Effort = A * Size^B * M

  where A and B are constants that depend on the sub-model, Size is the software size in SLOC or FP, and M is the product of the cost drivers and scaling factors.
- COCOMO II also provides formulas to estimate the development time (in months) and the number of developers (in persons) for each sub-model, based on the estimated effort and some other parameters.
- COCOMO II is a widely used and well-tested model that can provide accurate and reliable estimates for software projects of different types and sizes. It can also be calibrated and tailored to fit the specific needs and characteristics of an organization or a project .

A Parametric Productivity Model in SPM

- SPM stands for Software Project Management, which is the discipline of planning, organizing, monitoring, and controlling software projects.
- A parametric productivity model is a mathematical equation that relates the output of a software project (such as size, functionality, quality, etc.) to the input factors (such as effort, time, resources, etc.).
- A parametric productivity model can be used to estimate, measure, and improve the productivity of software projects, as well as to compare the performance of different projects, teams, or methods.
- A parametric productivity model typically has the following form:

```
Output = f(Input1, Input2, ..., Inputn)
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- Where Output is the dependent variable that represents the software project output, f is the function that describes the relationship between the output and the inputs, and Input1, Input2, ..., Inputn are the independent variables that represent the software project input factors.
- A parametric productivity model can be derived from empirical data, such as historical project data, expert judgment, or industry benchmarks, using statistical techniques, such as regression analysis, curve fitting, or machine learning.
- A parametric productivity model can be validated by testing its accuracy, reliability, and applicability on new or independent data sets, using metrics, such as mean absolute error, coefficient of determination, or prediction interval.
- A parametric productivity model can be calibrated by adjusting its parameters, such as coefficients, constants, or exponents, to fit the specific characteristics of a software project, such as domain, technology, or complexity.
- A parametric productivity model can be updated by incorporating new data, feedback, or knowledge, to reflect the changes in the software project environment, such as requirements, scope, or quality.
- A parametric productivity model can be applied by using its equation, or a tool that implements it, to estimate, measure, or improve the software project output, given the input factors, or vice versa.

Unit - 3 - Activity Planning and Risk Management

Activity Planning

- Activity planning is the process of identifying and organizing the tasks and resources needed to complete a software project within the given constraints of time, budget and quality.
- Activity planning involves the following steps:
  - Define the project scope and objectives
  - Identify the project deliverables and milestones

  - Decompose the project into work packages and tasks
  - Estimate the effort, duration and cost of each task
  - Assign resources and responsibilities to each task
  - Create a project schedule and a work breakdown structure (WBS)
  - Monitor and control the project progress and performance

Risk Management

- Risk management is the process of identifying, analyzing, prioritizing and mitigating the uncertainties and threats that may affect the software project objectives, deliverables and outcomes.
- Risk management involves the following steps:
  - Identify the potential risks and their sources, impacts and probabilities
  - Analyze the risks and their interdependencies, dependencies and correlations
  - Prioritize the risks based on their severity, likelihood and exposure
  - Plan the risk responses and contingency plans for each risk
  - Implement the risk responses and monitor the risk status and triggers
  - Review and update the risk register and the risk management plan

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Objectives of Activity Planning in SPM

- Activity planning is one of the most important management activities in software project management (SPM).
- Activity planning aims to provide a framework that enables the manager to make reasonable estimates of resources, cost, and schedule for the project.
- The objectives of activity planning are:
  - To identify the activities that need to be performed to complete the project successfully.
  - To determine the dependencies and sequence of the activities.
  - To estimate the duration, effort, and cost of each activity.
  - To allocate the resources and assign the responsibilities for each activity.
  - To produce a detailed schedule that shows the start and finish dates of each activity and the milestones of the project.
  - To produce a detailed plan that can be used to monitor and control the progress and performance of the project.
  - To produce a timed cash flow forecast that shows the expected income and expenditure of the project.

  - To identify the risks and uncertainties associated with the project and plan for their mitigation and contingency.

- Activity planning can be done using different approaches, such as:
  - The activity-based approach, which focuses on the tasks and deliverables of the project and uses a work breakdown structure (WBS) to decompose the project into manageable units.
  - The product-based approach, which focuses on the features and functions of the software product and uses a product breakdown structure (PBS) to decompose the product into components and subcomponents.
  - The hybrid approach, which combines the activity-based and product-based approaches and uses a combination of WBS and PBS to decompose the project and the product.

- Activity planning can also use different techniques, such as:
  - Network planning models, which use graphical representations of the activities and their dependencies, such as activity-on-node (AON) or activity-on-arrow (AOA) diagrams.
  - Critical path method (CPM), which identifies the longest path of dependent activities in the network and determines the minimum time required to complete the project.
  - Program evaluation and review technique (PERT), which considers the variability and uncertainty of the activity durations and calculates the expected time and the probability of completing the project within a given time.

Project Schedules in SPM

- Project schedules are the tools that communicate what work needs to be performed, which resources of the organization will perform the work and the timeframes in which that work needs to be performed.
- Project schedules help the project team to plan, execute and control the project activities and track the project progress.
- Project schedules are based on the project scope, work breakdown structure (WBS), resource availability, activity durations, dependencies and constraints.
- Project schedules can be presented in various formats, such as Gantt charts, network diagrams, milestone charts, etc.
- Project schedules are dynamic and need to be updated and revised throughout the project life cycle to reflect the changes in the project scope, resources, risks and issues.
- Project schedules are one of the key project management documents that are used to monitor and control the project performance, quality, cost and scope.

Activities in SPM

SPM stands for Software Project Management, which is the discipline of planning, organizing, directing, and controlling the activities of a software project. Some of the main activities involved in SPM are:

- **Project initiation**: This is the first phase of SPM, where the project scope, objectives, feasibility, risks, stakeholders, and deliverables are defined and agreed upon. The project charter, which is a document that summarizes the project's purpose, scope, and goals, is also created in this phase.
- **Project planning**: This is the second phase of SPM, where the project activities, resources, schedule, budget, quality, and communication are planned and documented. The project plan, which is a document that describes how the project will be executed, monitored, and controlled, is also created in this phase.
- **Project execution**: This is the third phase of SPM, where the project team performs the tasks and produces the deliverables according to the project plan. The project manager monitors and controls the project progress, quality, costs, and risks, and communicates with the stakeholders regularly.
- **Project closure**: This is the final phase of SPM, where the project is formally completed and delivered to the customer. The project manager evaluates the project performance, outcomes, and lessons learned, and documents the project closure report, which is a document that summarizes the project achievements, challenges, and recommendations. The project manager also releases the project resources and celebrates the project success with the team and the stakeholders.

Sequencing and Scheduling in SPM

- SPM stands for Software Project Management, which is the discipline of planning, organizing, executing, and controlling software projects.
- Sequencing and scheduling are two important aspects of SPM, as they help to determine the order and timing of the project activities and tasks.
- Sequencing is the process of determining which activities must be completed first, second, and so on, based on their dependencies, priorities, and constraints .
- Scheduling is the process of assigning resources, such as people, equipment, and budget, to the sequenced activities and estimating their duration, start, and finish dates .
- Sequencing and scheduling are interrelated, as the sequence of activities affects the schedule, and the schedule may require adjustments to the sequence.
- Sequencing and scheduling can be done using various techniques, such as:
  - Gantt charts, which are graphical representations of the project

schedule, showing the start and finish dates of each activity and their dependencies.
  - Network diagrams, which are graphical representations of the project activities and their logical relationships, such as precedence, parallelism, and feedback .
  - Critical path method (CPM), which is a technique that identifies the longest path of dependent activities in a network diagram and calculates the minimum time required to complete the project .
  - Program evaluation and review technique (PERT), which is a technique that estimates the duration of each activity based on optimistic, pessimistic, and most likely scenarios and calculates the expected time and variance of the project .
  - Resource leveling, which is a technique that adjusts the start and finish dates of the activities to balance the demand and availability of the resources.
- Sequencing and scheduling are essential for effective SPM, as they help to:
  - Define the scope and deliverables of the project
  - Establish the milestones and deadlines of the project
  - Allocate and optimize the use of the resources
  - Monitor and control the progress and performance of the project
  - Identify and manage the risks and uncertainties of the project
  - Communicate and coordinate with the stakeholders of the project

Network Planning Models in SPM

- Network planning models are used to plan and manage software projects by using graphical representations of activities and events.
- They help to visualize the sequence, duration, order, and dependencies of tasks necessary to complete the project.
- They also help to estimate the project cost, time, and resources, and to identify the critical path and the risks involved.
- There are two main types of network planning models: activity-on-node (AON) and activity-on-arrow (AOA).
- In AON, the activities are represented by nodes (boxes) and the dependencies are represented by arrows (lines) between the nodes.
- In AOA, the activities are represented by arrows and the nodes represent the start and end points of the activities.
- Both AON and AOA use the same notation to indicate the activity name, duration, and slack time.
- The activity name is written above the node or arrow, the duration is written below the node or arrow, and the slack time is written in brackets next to the node or arrow.
- The slack time is the amount of time that an activity can be delayed without affecting the project completion time.
- The critical path is the longest path of activities in the network, which

determines the minimum project completion time.
- The activities on the critical path have zero slack time and are called critical activities.
- Any delay in a critical activity will delay the whole project.
- The critical path can be identified by using the forward pass and backward pass methods, which calculate the earliest start time (EST), earliest finish time (EFT), latest start time (LST), and latest finish time (LFT) of each activity.
- The EST and EFT are calculated by adding the activity duration to the maximum of the EFTs of the preceding activities.
- The LST and LFT are calculated by subtracting the activity duration from the minimum of the LSTs of the succeeding activities.
- The slack time is then calculated by subtracting the EST from the LST or the EFT from the LFT.
- The critical path is the path of activities that have zero slack time.
- An example of a network planning model using AON notation is shown below:

AON example

- The critical path is A-B-D-E-F-G, with a total duration of 28 days.
- The slack time of each activity is shown in brackets next to the node.
- For example, activity C has a slack time of 4 days, which means it can be delayed by up to 4 days without affecting the project completion time.

Formulating Network Model in SPM

- SPM stands for Software Project Management, which is the process of planning, organizing, executing, and controlling software projects.
- A network model is a graphical representation of the activities and events involved in a software project, showing their sequence, duration, and dependencies.
- Formulating a network model in SPM involves the following steps:
  - Identify the project scope, objectives, deliverables, and stakeholders.
  - Define the work breakdown structure (WBS), which is a hierarchical decomposition of the project into manageable tasks.
  - Estimate the effort, cost, and time required for each task, using techniques such as expert judgment, analogy, parametric, or bottom-up.
  - Determine the logical order and precedence of the tasks, using techniques such as dependency analysis, critical path method, or PERT.
  - Draw the network diagram, which is a visual representation of the tasks and their dependencies, using symbols such as nodes, arrows, or boxes.

  - Analyze the network diagram, which involves identifying the critical path, slack time, float, and milestones.
  - Monitor and control the network model, which involves updating the progress, status, and changes of the tasks, using techniques such as earned value management, variance analysis, or risk management.
- A network model in SPM can help to:
  - Visualize the project scope, schedule, and resources.
  - Identify the critical activities and potential risks.
  - Optimize the project performance and quality.
  - Communicate the project plan and progress to the stakeholders.

Forward Pass & Backward Pass Techniques in SPM

- SPM stands for Software Project Management, which is the process of planning, organizing, executing, monitoring and controlling software projects.
- One of the tools used in SPM is the network diagram, which is a graphical representation of the activities and dependencies in a project.
- A network diagram consists of nodes (representing activities) and arrows (representing dependencies or precedence relationships).
- A network diagram can help to identify the critical path, which is the longest sequence of activities that determines the minimum project duration.
- A network diagram can also help to calculate the early and late start and finish dates of each activity, as well as the float or slack time, which is the amount of time an activity can be delayed without affecting the project duration.
- To calculate the early and late start and finish dates, and the float or slack time, two techniques are used: forward pass and backward pass.
- Forward pass is a technique to move forward through a network diagram, starting from the first activity, to determine the early start (ES) and early finish (EF) dates of each activity.
- ES is the earliest possible date an activity can start, given the dependencies and the project start date.
- EF is the earliest possible date an activity can finish, given the dependencies and the activity duration.
- The formula for ES and EF are:

  - ES = max(EF of all predecessors) or project start date if no predecessors
  - EF = ES + activity duration
- Backward pass is a technique to move backward through a network diagram, starting from the last activity, to determine the late start (LS) and late finish (LF) dates of each activity.
- LS is the latest possible date an activity can start, without delaying the project finish date.
- LF is the latest possible date an activity can finish, without delaying the project finish date.
- The formula for LS and LF are:
  - LF = min(LS of all successors) or project finish date if no successors
  - LS = LF - activity duration
- The float or slack time of an activity is the difference between its early and late start dates, or between its early and late finish dates. It can be calculated as:
  - Float = LS - ES or LF - EF
- An activity with zero float is on the critical path, meaning it cannot be delayed without affecting the project duration.
- An activity with positive float is not on the critical path, meaning it can be delayed up to the float amount without affecting the project duration.
- An activity with negative float is behind schedule, meaning it needs to be expedited to meet the project deadline.

Critical Path Method (CPM) in SPM

- Critical Path Method (CPM) is a technique where you identify tasks that are necessary for project completion and determine scheduling flexibilities .
- SPM stands for System Project Management, which is a discipline that focuses on managing complex systems and projects.
- CPM helps you to find the critical path in project management, which is the longest sequence of activities that must be finished on time to complete the entire project .
- CPM involves six steps:
  - Step 1: Specify Each Activity. List all the tasks or activities that are required to complete the project.
  - Step 2: Establish Dependencies (Activity Sequence). Identify the logical relationships and dependencies among the activities. For ex-

ample, some activities may need to be done before others, or some activities may be done in parallel.
  - Step 3: Draw the Network Diagram. Represent the activities and dependencies as a network diagram, using nodes to represent activities and arrows to represent dependencies. The network diagram shows the flow of the project from start to finish.
  - Step 4: Estimate Activity Completion Time. Assign a duration or time estimate to each activity, based on the resources, skills, and complexity involved.
  - Step 5: Identify the Critical Path. Calculate the earliest start and finish times and the latest start and finish times for each activity, using the forward pass and backward pass techniques. The critical path is the path with the longest duration, which determines the minimum time needed to complete the project. The activities on the critical path have zero slack or float, which means they cannot be delayed without affecting the project completion time.
  - Step 6: Update the Critical Path Diagram to Show Progress. Monitor and control the project progress by comparing the actual start and finish times of the activities with the planned start and finish times. Identify any deviations or delays and take corrective actions if needed. Update the critical path diagram to reflect the current status of the project.
- CPM is a useful tool for planning, scheduling, and managing projects, as it helps you to optimize the use of resources, identify potential risks and bottlenecks, and improve the quality and efficiency of the project outcomes .

Risk Identification in SPM

- Risk identification is the process of finding and documenting the possible sources of uncertainty that could affect the objectives of a software project .
- Risk identification is one of the most important and initial steps in risk management process, as it helps to prepare for the potential threats and opportunities that may arise during the project lifecycle .
- Risk identification is an iterative process, as new risks may be identified as the project progresses, old risks may become irrelevant, and the characteristics of existing risks may change .
- Risk identification involves brainstorming activities, where all the stakeholders meet together and generate ideas about the possible risks that could affect the project .
- Risk identification also involves the preparation of a risk list, which is a document that records the identified risks, their sources, categories, im-

pacts, probabilities, and other relevant information .

- Risk identification can use various techniques, such as:
  - Checklists: A list of common risks that have occurred in similar projects or domains, which can be used as a reference to identify risks for the current project .
  - Interviews: A method of asking questions to the project team, experts, customers, or other stakeholders, to elicit their opinions and experiences about the potential risks that could affect the project .
  - Surveys: A method of collecting data from a large number of respondents, using questionnaires or polls, to gather information about the possible risks that could impact the project .
  - SWOT analysis: A technique of analyzing the strengths, weaknesses, opportunities, and threats of the project, to identify the internal and external factors that could pose risks or create opportunities for the project .
  - Brainstorming: A group discussion technique where all the stakeholders meet together and generate ideas about the possible risks that could affect the project, without criticizing or evaluating them .
  - Delphi technique: A method of reaching a consensus among a group of experts, using a series of anonymous questionnaires, to identify and prioritize the risks that could influence the project .
  - Cause and effect analysis: A technique of identifying the root causes of the problems or issues that could affect the project, using diagrams or charts, to trace the effects of the risks to their sources .
  - Scenario analysis: A technique of creating and analyzing different possible situations or outcomes that could occur in the project, based on the assumptions and variables that could change .
  - Risk breakdown structure: A tool that organizes the identified risks into a hierarchical structure, based on their categories, sources, or impacts, to facilitate the analysis and management of the risks .
- Risk identification is a crucial step in risk management, as it helps to:
  - Increase the awareness and understanding of the project team and stakeholders about the potential risks that could affect the project .
  - Reduce the uncertainty and ambiguity of the project objectives, scope, schedule, budget, and quality .
  - Enhance the planning and decision making process of the project, by considering the possible impacts and consequences of the risks .
  - Improve the communication and collaboration among the project team and stakeholders, by involving them in the risk identification process and sharing the risk information .
  - Prepare for the mitigation and contingency actions, by prioritizing the risks and allocating the resources accordingly .
  - Increase the chances of achieving the project success, by avoiding or minimizing the negative effects of the risks and exploiting or maxi-

mizing the positive effects of the opportunities .

Risk Assessment in SPM

- SPM stands for Single Point Mooring, which is a floating buoy that serves as a loading and offloading point for oil tankers in offshore locations.
- Risk assessment is the process of identifying, analyzing, and evaluating the potential hazards and consequences of an activity or event.
- Risk assessment in SPM is important to ensure the safety and reliability of the subsea pipelines that connect the SPM to the shore or to the production platform.
- The main risks associated with SPM are:
  - Third party interference, such as dropped or dragged anchors, ship collisions, or fishing activities, that can damage the subsea pipelines or the SPM itself.
  - Environmental factors, such as waves, currents, winds, corrosion, or marine growth, that can affect the structural integrity and performance of the subsea pipelines or the SPM.
  - Operational factors, such as human error, equipment failure, or improper maintenance, that can cause leaks, spills, or accidents.
- The main steps of risk assessment in SPM are:
  - Define the scope and objectives of the risk assessment, such as the system boundaries, the risk criteria, and the stakeholders involved.
  - Identify the potential hazards and threats that can affect the SPM and the subsea pipelines, using historical data, expert judgment, or other sources.
  - Analyze the probability and impact of each hazard or threat, using quantitative or qualitative methods, such as fault tree analysis, event tree analysis, or risk matrix.
  - Evaluate the risk level of each hazard or threat, comparing it with the risk criteria and ranking it according to its severity and likelihood.
  - Identify and implement the risk mitigation measures, such as design changes, operational procedures, or contingency plans, that can reduce the risk level to an acceptable level.
  - Monitor and review the risk assessment, updating it as new information or changes occur, and verifying the effectiveness of the risk mitigation measures.
- Some of the standards and guidelines that can be used for risk assessment in SPM are:
  - DNVGL RP F107: Risk assessment of pipeline protection, which provides a methodology and criteria for assessing the risk of subsea pipelines due to third party interference and environmental factors.
  - UNSMS Security Policy Manual: Policy on Security Risk Management, which provides a framework and process for managing security risks in the United Nations system.

  - ISO 31000: Risk management – Principles and guidelines, which provides a generic and comprehensive approach for risk management in any context or organization.

Risk Planning in SPM

- Risk planning is the process of identifying, analyzing, and managing the potential risks that may affect a software project.
- Risk planning is one of the core activities of software project management (SPM), which aims to deliver software products that meet the customer's requirements, budget, and schedule.
- Risk planning involves the following steps:
  - **Risk identification**: This is the process of finding out the possible sources of uncertainty that may affect the project's objectives, such as technical, organizational, environmental, or external factors. Some techniques for risk identification are brainstorming, checklists, interviews, surveys, and historical data analysis .
  - **Risk analysis**: This is the process of assessing the probability and impact of each identified risk, and prioritizing them according to their severity and urgency. Some techniques for risk analysis are qualitative methods (such as risk matrices, risk ranking, and risk categorization) and quantitative methods (such as expected value, decision trees, and Monte Carlo simulation) .
  - **Risk response planning**: This is the process of developing strategies and actions to reduce, avoid, transfer, or accept the risks, depending on their nature and level. Some techniques for risk response planning are contingency plans, fallback plans, risk reserves, risk owners, and risk triggers .
  - **Risk monitoring and control**: This is the process of tracking, reviewing, and updating the risk status, and implementing the risk response plans as needed. Some techniques for risk monitoring and control are risk audits, risk reports, risk reviews, and risk metrics .
- Risk planning is an iterative and ongoing process throughout the project life cycle, as new risks may emerge or existing risks may change over time.
- Risk planning is a collaborative and participatory process that involves the project team, stakeholders, and experts, as well as tools and techniques to support the risk management activities.
- Risk planning is a proactive and preventive process that aims to minimize the negative effects of risks and maximize the positive opportunities that may arise from them.
- Risk planning is a critical and beneficial process that can improve the

project performance, quality, and success.

Risk Management in SPM

- Risk management is a sequence of steps that help a software team to understand, analyze and manage uncertainty.
- Risk management is one of the core tents of the philosophy of SPM, as it helps to ensure the successful delivery of software projects within budget, time and quality constraints.
- The risk management process consists of the following steps :
  - Risk identification: The process of identifying the potential sources of risk that may affect the software project, such as technical, organizational, operational, environmental, legal, or market factors.
  - Risk analysis: The process of assessing the probability and impact of each identified risk, and prioritizing them according to their severity and urgency.
  - Risk planning: The process of developing strategies to avoid, reduce, transfer, or accept the risks, and allocating resources and responsibilities for implementing them.
  - Risk monitoring: The process of tracking the status of the risks and the effectiveness of the risk management strategies, and updating the risk register and the risk plan as needed.
- The risk management process requires the use of various tools and techniques, such as risk register, risk matrix, risk breakdown structure, risk exposure, risk mitigation, risk contingency, risk review, and risk audit .
- The risk management process is an iterative and continuous activity that runs in parallel to the other project management activities throughout the software project lifecycle.
- The risk management process is a shared responsibility of the software project manager, the software team, and the stakeholders, who should communicate and collaborate effectively to identify, analyze, plan, and monitor the risks .
- The risk management process is a key component of the software quality assurance, as it helps to ensure that the software meets the requirements and expectations of the customers and the users, and that the software is delivered with minimal defects and errors .
- The risk management process is also applicable to other domains and contexts, such as security, safety, and strategic portfolio management, where it helps to analyze and manage the safety and security risks to personnel, assets, and operations, and to align the strategy to the work and the outcomes .

PERT Technique in SPM

- PERT stands for Program Evaluation and Review Technique. It is a statistical tool used in project management to analyze and represent the tasks involved in completing a given project.
- PERT was first developed by the US Navy in 1958 during the Polaris missile development program and then applied to other industries .
- PERT is used to identify task dependencies and critical paths, plan resources, estimate task duration, and identify potential risks .
- PERT helps to define and sequence activities, coordinate resources, and track progress .
- PERT uses a network diagram to represent the project activities and their interrelationships. Each activity is represented by a node and each dependency is represented by an arrow .
- PERT uses a three-point estimating technique to calculate the expected duration of each activity. The three points are the optimistic, pessimistic, and most likely estimates. The expected duration is the weighted average of these three estimates .
- PERT uses the expected durations and the network diagram to calculate the earliest and latest start and finish times of each activity. These times are used to determine the critical path, which is the longest path in the network diagram. The critical path indicates the minimum time required to complete the project and the activities that have no slack or float .
- PERT also uses the expected durations and the network diagram to calculate the variance and standard deviation of each activity and the project as a whole. These measures are used to assess the probability of completing the project within a given time frame and the level of uncertainty or risk involved .
- PERT is a useful technique for planning and controlling complex and uncertain projects. It helps to identify the critical activities, allocate resources, monitor progress, and evaluate performance .

Monte Carlo Simulation in SPM

- Monte Carlo simulation is a mathematical technique that uses random sampling to estimate the possible outcomes of an uncertain event.
- SPM stands for service parts management, which is the process of planning, forecasting, and optimizing the inventory and distribution of spare parts for products or equipment.
- Monte Carlo simulation can be helpful for SPM in the following ways:
  - It can evaluate the impact of demand variability, lead time variability, and supply disruptions on the inventory and service levels of service parts.
  - It can compare different stocking policies and strategies to find the optimal trade-off between inventory cost and service level.

  - It can account for the dependencies and correlations among different service parts, locations, and customers.
  - It can provide a probabilistic distribution of the outcomes, rather than a single point estimate, which can help in risk analysis and decision making .
- To perform a Monte Carlo simulation for SPM, the following steps are required:
  - Define the problem and the objective of the simulation.
  - Identify the input variables and their probability distributions, such as demand, lead time, supply availability, etc.
  - Generate random samples from the input distributions using a random number generator.
  - Run the simulation model using the random samples as inputs, and record the output variables, such as inventory level, service level, fill rate, etc.
  - Repeat the simulation for a large number of times (e.g., 10,000) to obtain a large sample of output values.
  - Analyze the output values using descriptive statistics, such as mean, standard deviation, confidence intervals, etc., or graphical methods, such as histograms, scatter plots, etc.
  - Interpret the results and draw conclusions based on the objective of the simulation.

Resource Allocation in SPM

- Resource allocation in SPM (software project management) is the process of allocating, sequencing, and managing resources for a given project.
- Resources include staff, budgets, materials, and equipment needed for the successful completion of a project.
- Resource allocation in SPM is important because it gives a clear picture of the amount of work that has to be done, the time and cost required, and the progress and performance of the project team.
- Resource allocation in SPM involves the following steps:
  - Divide the project into tasks and assign them to different phases or stages of the project lifecycle.
  - Estimate the effort, duration, and cost of each task and the dependencies and constraints among them.
  - Identify the available resources and their skills, availability, and capacity.
  - Assign the resources to the tasks based on their suitability, priority, and availability.
  - Monitor and control the resource utilization and allocation throughout the project and make adjustments as needed.
  - Evaluate the resource allocation and its impact on the project outcomes and benefits.

- Resource allocation in SPM can be done using various techniques and tools, such as:
  - Resource breakdown structure (RBS): a hierarchical representation of the resources by type, category, and subcategory.
  - Resource histogram: a graphical display of the resource availability and demand over time.
  - Resource leveling: a technique to balance the resource demand and supply by adjusting the task start and finish dates or durations.
  - Resource smoothing: a technique to minimize the fluctuations in resource demand by using slack or float time.
  - Resource allocation matrix (RAM): a table that shows the relationship between the resources and the tasks or deliverables.
  - Resource management software: a software application that helps to plan, schedule, track, and optimize the resource allocation and utilization.

Cost Schedules in SPM

- Cost schedules are detailed and accurate estimates of the project costs over time, showing weekly or monthly costs for each activity, resource, and deliverable.
- Cost schedules are used to:
  - Analyze and forecast the project's total costs and budget allocation .
  - Monitor and control the project's actual costs vs. planned costs and identify any cost variances .
  - Communicate the project's cost performance to stakeholders and obtain approval for any changes .
- Cost schedules are derived from:
  - The work breakdown structure (WBS), which defines the project's scope and deliverables.
  - The effort and duration estimations for each element of the WBS.
  - The cost categories and rates for each resource and activity.
  - The project schedule, which determines the start and end dates for each activity and milestone .
- Cost schedules are updated and revised throughout the project life cycle, as more information becomes available and changes occur.
- Cost schedules are part of the cost management plan, which outlines the processes and tools for estimating, budgeting, and controlling costs.